Syed Muhammad Taha Hassan Shah

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Professional Summary

Sophomore Software Engineering student at UBIT – University of Karachi with hands-on experience in machine learning, full-stack web development, and cloud-native deployments. Proficient in Python, JavaScript, TypeScript, and C++; skilled in scikit-learn, TensorFlow, PyTorch, and AWS services. Ranked in the top 12% (138th of 1,142 teams) at the NeurIPS 2025 × Kaggle Google Code Golf Championship, competing against PhD-level researchers and industry veterans. Seeking a machine learning engineering, AI development, or software engineering role where strong analytical and technical skills can drive measurable impact.

Education

UBIT – University of Karachi — Karachi, Pakistan
Bachelor of Science in Software Engineering — *2025 – 2028*

Bahria College Karsaz — Karachi, Pakistan
Intermediate in Pre-Engineering – Percentage: 72.5% — *2023 – 2025*

Crescent Bahria Cadet College — Karachi, Pakistan
Matriculation in Computer Science – Percentage: 88.8% — *2020 – 2022*

Experience

Freelance Machine Learning Engineer — 2025 – Present
Self-Directed ML Projects | Remote — *Karachi, Pakistan*

- Designed and deployed end-to-end ML pipelines using Python, scikit-learn, Pandas, NumPy, and Jupyter covering data ingestion, preprocessing, feature engineering, and model evaluation.
- Performed exploratory data analysis (EDA) and applied feature scaling, dimensionality reduction, and cross-validation techniques on real-world datasets to improve model accuracy.
- Deployed production-ready ML models as interactive web applications using Streamlit, enabling live inference and real-time data visualization for non-technical users.
- Developed deep learning models using TensorFlow and PyTorch, exploring supervised learning, neural network architectures, and optimization strategies.
- Maintained all projects with Git/GitHub, applying modular architecture and clean code principles; AWS (EC2, S3, Lambda) used for cloud-based deployments.

Projects

Cardio Risk AI | *Python, scikit-learn, Streamlit, Pandas, NumPy*

- Built a cardiovascular risk assessment system using KNN (K-Nearest Neighbors) machine learning model to predict heart disease risk based on patient health data.
- Deployed the model as an interactive web application using Streamlit, allowing users to input health parameters and receive real-time risk predictions.

Significo-Clone | *HTML, CSS, JavaScript, AWS S3, AWS CodePipeline*

- Built a pixel-perfect static clone of the Significo healthcare website using pure HTML, CSS, and vanilla JavaScript with smooth scrolling and interactive elements.
- Deployed as a fully automated static site on AWS S3 with CI/CD pipeline using AWS CodePipeline + GitHub webhook for zero-downtime updates.

Derivium | *HTML, CSS, JavaScript, Python, Flask, SymPy, Plotly.js, GSAP, KaTeX*

- Built a full-stack calculus visualization web app enabling users to compute limits, derivatives, and partial derivatives with symbolic mathematics backend.
- Integrated interactive 2D/3D graph plotting using Plotly.js with smooth GSAP animations and LaTeX-quality math rendering via KaTeX.

Competitions & Awards

NeurIPS 2025 x Kaggle - Google Code Golf Championship *Aug – Oct 2025*
Hosted by Google | Remote *Final Rank: 138 / 1,142 Teams (Top 12%)*

- Competed against 1,142 teams worldwide including PhD-level researchers, industry veterans, and specialists with 40+ years of experience as an undergraduate student.
- Achieved a peak rank of 109 and a final rank of 138; developed multiple submission versions using byte-level minimization, creative hacks, and unconventional algorithmic logic under competitive time pressure.
- Submissions were incorporated into Google's training dataset for Gemini AI contributing production-quality optimization code that informed the development of a frontier large language model.
- Open-sourced all submissions and experimental approaches on GitHub to help future competitors learn optimization strategies and contribute back to the AI/ML community.

Leadership & Volunteering

AWS Student Builder Group | Core Team Member *April 2026 – Present*
Bahria University Karachi Campus *Karachi, Pakistan*

- Organize cloud computing workshops, technical sessions, and community events focused on hands-on AWS skill-building for students.
- Lead cross-team coordination and end-to-end execution of club operations, supporting onboarding of new members into cloud-first learning pathways.

Technical Skills

Languages: Python, JavaScript, TypeScript, C++, HTML, CSS
ML & Data Science: scikit-learn, TensorFlow, PyTorch, NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook
Frontend: React, Next.js, Tailwind CSS, GSAP, Plotly.js, KaTeX, REST APIs, Flask
Cloud & DevOps: AWS (S3, EC2, Lambda, RDS, DynamoDB, SQS, SNS, Redshift, CloudWatch, IAM, EventBridge, CodePipeline), Vercel, Git, GitHub, CI/CD Pipelines
Developer Tools: VS Code, PyCharm, Streamlit, Jupyter Notebook